Drones in Agriculture: Bibliometric State of the Art and Growth Forecasts to 2035

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Abstract—**Unmanned aerial vehicles (UAVs) are promoted as a cornerstone of precision agriculture, both as sensing platforms and, increasingly, as actuators for spraying, yet the pace and the likely saturation of this adoption have not been quantified. We compiled annual counts of peer-reviewed articles combining UAV terms with agricultural terms from the open bibliographic index OpenAlex for 2005 to 2025 (18,506 articles after noise filtering), together with four normalising denominators and seven application sub-domains, and fitted four growth models (exponential, logistic, Gompertz and Bass diffusion) by Poisson deviance in a toolbox-free MATLAB pipeline. Models were compared with a quasi-likelihood information criterion, validated by rolling-origin backtesting and bootstrapped for forecast intervals. Output grew at 28.5 percent per year (29 articles in 2005, 4,381 in 2025), with a structural break in 2009. No model dominates for absolute output: the model average predicts 8,174 articles in 2028 and 40,260 in 2035, with a ten-year interval of 13,390 to 55,050. The share of the agricultural literature that uses UAVs is saturating later than in neighbouring fields: a logistic model with weight 0.91 gives a plateau of 13.8 articles per 1,000, of which 62 percent was reached in 2025. Growth is fastest in yield prediction, irrigation and disease detection, while spraying makes the UAV an actuator with distinct regulatory constraints. Implications for UAV design and autonomy are discussed.**

Keywords—unmanned aerial vehicles, precision agriculture, agricultural drones, bibliometrics, technology forecasting, growth models

I. Introduction

A decade after the first reviews announced small unmanned aerial vehicles (UAVs, also UAS or RPAS) as the missing high-resolution, on-demand platform of precision agriculture [1], agricultural drones monitor crops, estimate yields, detect weeds and diseases, manage irrigation and, at scale in East Asia, spray pesticides and fertilisers [2], [3]. For the robotics community two questions remain open. First, how fast is the adoption of UAVs in agriculture actually growing once the growth of the whole agricultural literature is taken into account? Second, is that growth still exponential or approaching saturation, and what does the answer imply for where sensing, actuation, autonomy and regulatory effort should be directed over the next decade?

This paper answers both questions with a reproducible bibliometric time series and a transparent forecasting pipeline, applying the methodology we developed for UAVs in ecology and environmental monitoring to the agricultural domain. Our contributions are: (i) an openly queryable annual data set (2005–2025) of UAV research in agriculture, with a documented noise filter, four denominators and seven application sub-domains, built from the OpenAlex index [4]; (ii) a description of the current state of the field, including trend tests, a structural break and the distribution of output over sub-domains, countries and venues; (iii) a MATLAB algorithm, free of toolbox dependencies, that fits and compares four classical growth models, validates them out of sample and produces bootstrapped forecasts for 3, 5, 7 and 10-year horizons; and (iv) a discussion of what the forecasts mean for UAV engineers, including the actuation role that distinguishes agriculture from purely observational domains.

The paper is organised as follows. Section II reviews the evidence on UAV performance in agricultural applications and the growth-curve tradition in technology forecasting. Section III describes the data and the algorithm. Section IV reports the results and Section V discusses them. Section VI concludes.

II. Related Work

Reviews of UAVs in precision agriculture converge on a common inventory of applications: crop growth and health monitoring with visible, multispectral, hyperspectral and thermal sensors; high-throughput field phenotyping; weed mapping for site-specific management; irrigation scheduling from canopy temperature and vegetation indices; disease and pest detection; yield estimation; and spraying [1], [2], [3], [6]. Early work already demonstrated the key building blocks: a UAV configured for early site-specific weed management [5], and multi-sensor data fusion that predicts soybean yield within a field from UAV imagery using deep learning [7]. Field-based phenotyping has become a discipline of its own, with UAVs replacing ground vehicles as the primary carrier [8]. Two recurring engineering caveats temper the enthusiasm: radiometric calibration and the gap between vegetation indices and agronomic variables limit transferability across fields and seasons [2], [9], and the economics of UAV operations only close at sufficient field sizes or service-provider models [9]. Spraying is the application in which the UAV stops being a sensor and becomes an actuator: adoption is led by East Asia, with more than 120,000 plant-protection drones operating in China and roughly 62 million hectares treated in 2021, and about a third of Japanese rice fields sprayed from unmanned helicopters as early as 2016 [17];

reviews document rapid progress but also open problems in droplet drift, deposition uniformity and dose control [6].

Growth-curve forecasting of technology adoption has a long tradition. The Bass diffusion model [10] describes adoption per period as the sum of innovation and imitation effects; logistic and Gompertz curves describe levels approaching a ceiling; a 25-year review concluded that no single curve dominates and recommended combining models and reporting model uncertainty [11]. Publication counts are widely used as an adoption proxy, with the caveat that the scientific literature itself grows by several percent per year [12], so unnormalised counts overstate the growth of any topic. Multimodel inference with Akaike weights [13] provides a principled way to combine competing curves. We follow these recommendations: we fit four models, normalise by the growth of the field, weight the models by a quasi-likelihood information criterion, and report intervals that include model uncertainty.

III. Data and Methods

A. Bibliometric Corpus

Data were retrieved on 28 August 2026 from the OpenAlex REST API [4], aggregated by publication year (endpoint /works, group_by=publication_year) with document type restricted to articles and reviews. The core query D AND A NOT N searched titles and abstracts, where D = (drone OR drones OR UAV OR UAVs OR UAS OR "unmanned aerial" OR "unmanned aircraft" OR RPAS OR "remotely piloted aircraft"), A is a disjunction of 38 agricultural terms (agriculture, farm, crop, yield, irrigation, fertilizer, pesticide, spraying, weed, phenotyping, livestock, orchard, vineyard, "plant disease", "soil moisture" and their variants), and N is a noise filter added after a manual precision check: it removes the Drosophila GAL4/UAS genetics literature, apian "drones", and medical and institutional homonyms of the UAV acronyms. Four denominators were retrieved with identical filters: A alone (the agricultural literature searched in the same way, used as the primary denominator), the OpenAlex subfield Agronomy and Crop Science, the field Agricultural and Biological Sciences, and all indexed works. Seven overlapping sub-domain queries combined D with terms for crop monitoring and phenotyping, spraying, weed detection, irrigation and water stress, disease and pest detection, yield prediction, and livestock. Because OpenAlex is updated retroactively, the retrieval date is part of the data definition; the exact Boolean strings and a codebook accompany the MATLAB code.

The fitting window is 2005–2025 (n = 21 complete years); 2026 is incomplete and is excluded. The query is not perfectly precise. In a manual check of a random sample of 100 records of the unfiltered query (seed 20260828), coded with AI assistance and reviewed by the authors, 61 percent of the records surviving the noise filter [Wilson 95 percent CI 51, 70] were UAV studies with an agricultural application, a further 31 percent contained a UAV in a non-agricultural role, and 8 percent were residual noise. In an early-period sample (2005–2015, n = 35 after filtering) the relevant share was 37 percent [23, 54]; the difference is significant (two-proportion z = 2.38, p = 0.017), because the early literature is dominated by generic UAV engineering papers that mention agriculture only as a prospective application. Absolute counts therefore overstate strictly agricultural output by roughly 1.6 times, and the growth of that output is, if anything, faster than the growth of the query series. A stricter query that also excludes ecological and forestry terms grows at 28.5 percent per year as well, so the trend conclusions are robust. Finally, all series jump in 2025 (the agricultural denominator by 26 percent), which is partly an indexing artefact; normalisation attenuates it and a sensitivity analysis without 2025 is reported.

B. Descriptive Analysis

For the counts y_t and the share $s_t = 1000\ y_t/N_t$ (articles per 1,000 agricultural articles) we report the compound annual growth rate, the Mann–Kendall trend test with tie correction and Sen's slope with its 90 percent confidence interval [14], [15], and a one-breakpoint segmented log-linear regression found by grid search and tested against a single slope with a Chow-type F test (approximate, because the break is estimated).

C. Growth Models

Four models of the expected annual output $\mu(t)$ were fitted, with $\tau = t - 2005$:

$$\mu(t) = a \exp(b\,\tau) \tag{1}$$

$$\mu(t) = K / [1 + \exp(-r(\tau - t_m))] \tag{2}$$

$$\mu(t) = K \exp[-\exp(-r(\tau - t_m))] \tag{3}$$

$$\mu(t) = m\,[F(\tau+1) - F(\tau)] \tag{4}$$

where (1) is exponential growth without saturation, (2) the logistic and (3) the Gompertz curve with capacity K, rate r and inflection year t_m, and (4) the Bass diffusion model [10] with market potential m, coefficient of innovation p and coefficient of imitation q, where $F(x) = [1 - e^{-(p+q)x}] / [1 + (q/p)\, e^{-(p+q)x}]$ is the cumulative adoption fraction. For the share, the same curves describe the rate $g(t)$ with the denominator as an offset, $\mu_t = N_t\, g(t)$, so that share forecasts do not require a forecast of the denominator; the Bass model is omitted for the share because it describes adoptions per period, not a level.

D. Estimation, Model Selection and Uncertainty

Parameters were estimated by minimising the Poisson deviance

$$D = 2\,\Sigma_t\,[y_t \ln(y_t/\mu_t) - (y_t - \mu_t)] \tag{5}$$

with the Nelder–Mead simplex (MATLAB fminsearch) from multiple starting points and with positive parameters log-transformed. The deviance gives comparable weight to the small early and the large late years and yields a proper log-likelihood for information criteria. The counts are strongly overdispersed (Pearson dispersion $\hat{c} = 14.1$ for the

counts, 8.6 for the share), so models were compared with the small-sample quasi-likelihood criterion

$$\mathrm{QAIC}_c = -2 \ln L / \hat{c} + 2k + 2k(k+1)/(n-k-1) \qquad (6)$$

and combined with Akaike weights [13]. Out-of-sample accuracy was assessed by rolling-origin backtesting: each model was refitted on data up to 2018, 2020 and 2022 and evaluated on the remaining years by the mean absolute percentage error (MAPE) [16]. Forecast uncertainty was obtained by a parametric negative-binomial bootstrap: 1,000 replicate series per model, every model refitted on every replicate, QAIC_c weights recomputed and the weighted forecast recorded, giving 90 percent intervals that include both parameter and model uncertainty. Forecasts are reported for 2028, 2030, 2032 and 2035. The pipeline (ten MATLAB functions, no toolbox required, also executable in GNU Octave) was independently replicated in Python; parameter estimates and deviances agree to all reported digits.

IV. Results

A. Current State

Annual output rose from 29 articles in 2005 to 4,381 in 2025, a compound annual growth rate of 28.5 percent over the whole period and 30.4 percent over the last ten years; 18,506 articles were published in 2005–2025. The Mann–Kendall test is unambiguous ($S = 208$, $Z = 6.25$, $p = 4 \times 10^{-10}$), with a Sen slope of 125 articles per year [90 percent CI 86, 171]. The segmented regression locates a break in 2009: growth accelerated from 16 percent to 32 percent per year ($F = 5.2$, $p \approx 0.018$, approximate), coinciding with the arrival of low-cost multirotor platforms and consumer-grade multispectral cameras. Normalised by the agricultural literature, the share rose from 0.21 to 8.53 articles per 1,000 (Sen slope 0.38 per 1,000 per year [0.28, 0.47]); within the UAV literature as a whole, the agricultural fraction rose from 3.0 to 17.1 percent.

Output is concentrated: China (4,421 articles with at least one Chinese author in the unfiltered query), the United States (3,239), India (1,857), Germany (737) and the United Kingdom (690) lead 164 contributing countries; Croatia contributed 34 articles, all since 2013. The leading venues are Remote Sensing (1,026), Computers and Electronics in Agriculture (582), Drones (368), Agronomy (364) and Sensors (329); 72 percent of the corpus is open access and 942 articles have more than 99 citations. Fig. 1 shows the sub-domains. Over 2015–2025 yield prediction grew fastest (6 to 372 articles, 51 percent per year), followed by irrigation and water stress (42 percent), disease and pest detection (40 percent), crop monitoring and phenotyping (38 percent) and spraying (36 percent); the smallest and slowest sub-domains are weed detection (29 percent) and livestock (25 percent).

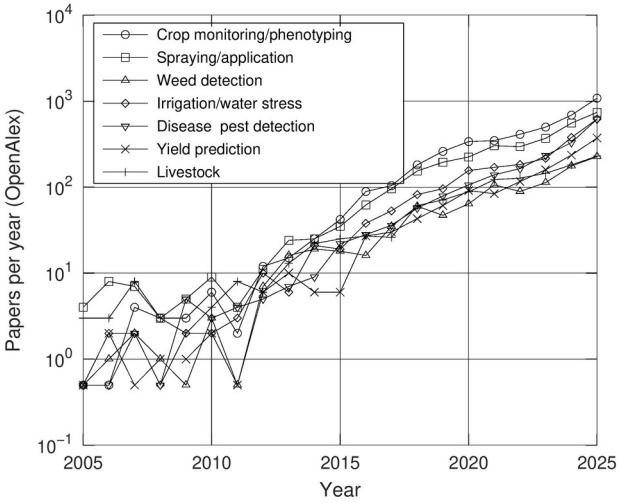


Figure 1. Annual number of OpenAlex articles combining UAV terms with seven agricultural sub-domains, 2005–2025 (logarithmic scale; the sub-domain queries overlap).

B. Growth Models for Annual Output

Table I compares the four models fitted to the annual counts. No model dominates: the QAIC_c weights range from 0.15 (Bass) to 0.39 (exponential), and, unlike in neighbouring observational domains, the exponential model receives the largest weight. The logistic model places the inflection point around 2030 with a capacity of about 21,000 articles per year — far above the current level — the Bass model peaks only in 2033, and the Gompertz fit degenerates towards an exponential shape. Backtesting gives MAPE of 28 to 34 percent over three origins, with the Gompertz model best from the 2022 origin (11 percent).

TABLE I. Growth Models Fitted to Annual Article Counts, 2005–2025

Model	k	Deviance	QAIC_c	Weight	Parameters
Exponential	2	283.4	36.1	0.391	$a = 26.6$, $b = 0.253$ (28.7 %/yr)
Logistic	3	269.4	37.9	0.163	$K = 21{,}030$, $r = 0.269$, $t_m = 2030.4$
Gompertz	3	253.0	36.7	0.292	$K = 3.2 \times 10^9$, $r = 0.017$, $t_m = 2181$
Bass	3	271.0	38.0	0.154	$m = 1.9 \times 10^5$, $p = 1.1 \times 10^{-4}$, $q = 0.267$

Fig. 2 shows the fits and the forecasts and Table II lists them. The models agree for the three-year horizon (7,260 to 8,850 articles in 2028; model average 8,174 [90 percent interval 6,232, 9,304]) and diverge afterwards: in 2035 the point forecasts range from 12,380 (Bass) to 51,810 (exponential), the model average is 40,260 and the interval spans 13,390 to 55,050. The sensitivity analysis shows the familiar dependence on the last observation: without 2025 the logistic capacity falls to about 5,300 and the Bass model predicts a decline, so a single year, partly an indexing artefact, again determines the estimated ceiling.

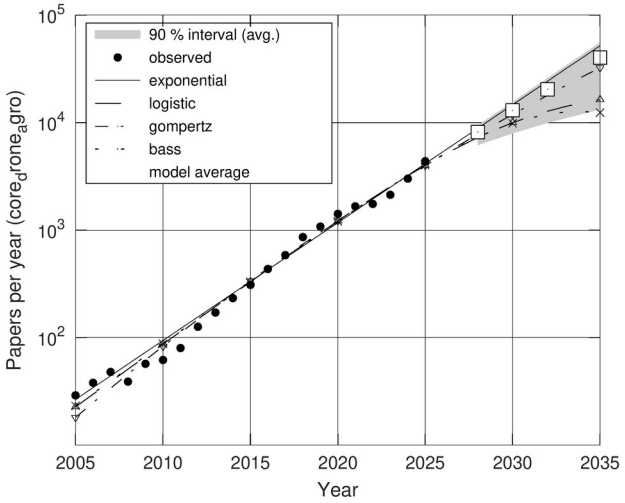


Figure 2. Annual OpenAlex articles combining UAV and agricultural terms (dots), four growth models fitted by Poisson deviance (lines) and the $QAIC_c$-weighted model average with its 90 percent negative-binomial bootstrap interval (squares, shaded) for 2028–2035.

TABLE II. FORECAST ANNUAL OUTPUT: POINT ESTIMATES AND 90 PERCENT BOOTSTRAP INTERVALS OF THE MODEL AVERAGE

Model	2028	2030	2032	2035
Exponential	8,850	14,660	24,290	51,810
Logistic	7,258	9,977	12,770	16,320
Gompertz	7,783	11,910	17,960	32,460
Bass	7,304	9,871	11,950	12,380
Model average	**8,174**	**13,000**	**20,440**	**40,260**
90 % interval	6,232–9,304	8,012–15,430	10,040–25,670	13,390–55,050

C. Share of the Agricultural Literature

The share (Fig. 3) again behaves more regularly than the counts. The logistic rate model receives most of the weight (0.91; the exponential model is rejected with a $QAIC_c$ difference of 18) and estimates a plateau of K = 13.8 articles per 1,000 agricultural articles with the inflection in 2023–2024. The share observed in 2025, 8.5 per 1,000, is 62 percent of that plateau — agriculture is thus visibly earlier on the adoption curve than ecology and environmental monitoring, where the same methodology finds 85 percent of the plateau already reached. The model-averaged forecast is 12.6 per 1,000 in 2028 [10.9, 14.6], 15.8 in 2030 [12.5, 20.3] and 26.4 in 2035 [16.0, 46.9], where the wide upper tail comes from the residual weight of the Gompertz curve. Backtesting from the 2022 origin gives a MAPE of 7.6 percent for the logistic model against 28.5 percent for the exponential one.

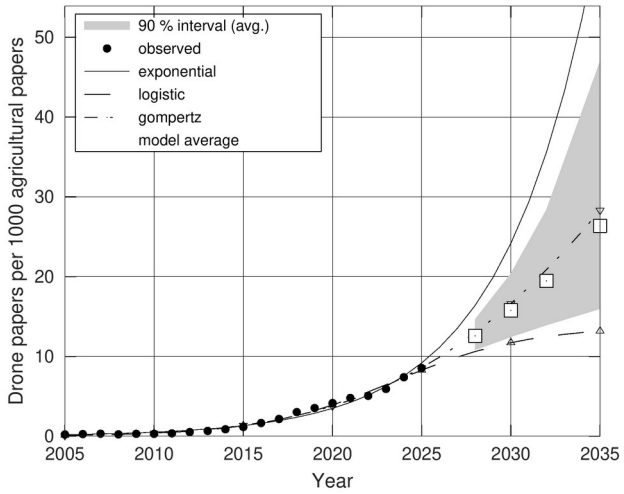


Figure 3. Share of UAV-related articles per 1,000 agricultural articles (OpenAlex), fitted exponential, logistic and Gompertz rate models with the agricultural literature as offset, and the $QAIC_c$-weighted forecast with its 90 percent bootstrap interval.

V. DISCUSSION

A. What the Forecasts Mean

Three findings frame the interpretation. First, the reliable forecast horizon for absolute output is again about three years; by 2035 the choice of curve dominates any statistical uncertainty. Second, the share analysis indicates that UAVs in agriculture are still in the steep part of the adoption curve: at 62 percent of the estimated plateau, with the inflection dated only 2023–2024, the field has more headroom than neighbouring observational domains. Third, the sub-domain ranking shows where that headroom lies: the fastest growth is in data-driven agronomy — yield prediction, irrigation scheduling and disease detection — rather than in the mapping applications that dominated the first decade.

B. Implications for the Robotics Community

Agriculture is the domain in which the UAV is becoming an actuator, not only a sensor, and this shifts the engineering agenda. Spraying at the scale documented in East Asia [17] makes payload capacity, droplet drift control, deposition uniformity, RTK-grade positioning and multi-vehicle coordination the binding constraints, together with dose-control loops that couple perception (prescription maps from the monitoring sub-domains) to actuation on the same platform or in sensor–sprayer teams [3], [6]. On the sensing side, the fastest-growing sub-domains need transferable models rather than new cameras: radiometric calibration, cross-season and cross-field generalisation of vegetation-index and deep-learning models remain the bottlenecks identified by the reviews [2], [9] and visible in our precision check, where model-transfer papers recur. Livestock and weed detection, the slowest sub-domains, are limited by economics and by ground-robot competition, respectively. Finally, the regulatory asymmetry deserves attention: in the European Union aerial spraying of pesticides is prohibited by Directive 2009/128/EC except under derogation [18], and UAV spraying operations additionally fall in the "specific" category of Regulation

(EU) 2019/947 [19], so the actuation applications that drive Asian adoption are largely closed in Europe; European growth therefore concentrates in sensing, which our European-country counts reflect. For European robotics, the near-term opportunity is monitoring and the preparation of the evidence base that regulators require for spraying derogations.

C. Limitations

Four limitations bound the conclusions. First, the corpus depends on one index and one query whose strict precision is 61 percent, and, unlike in our companion study of environmental UAVs, precision was significantly lower in the early period, so the earliest counts overstate agricultural activity; the noise filter removes the identifiable homonyms but not generic UAV engineering papers. Second, the 2025 observation is partly an indexing artefact and carries disproportionate weight in the saturating models. Third, the sub-domain queries overlap and describe relative growth, not a partition of the total. Fourth, model uncertainty is represented by four parametric shapes only; a regulatory change — for example a European opening towards spraying derogations — would not be captured by any of them. The adoption figures from industry and extension sources [17] are heterogeneous and are used for context only.

VI. Conclusion

A reproducible bibliometric series shows that research on UAVs in agriculture grew by 28.5 percent per year between 2005 and 2025, with a structural acceleration in 2009, and that its share of the agricultural literature is heading towards a plateau of about 14 articles per 1,000 of which less than two thirds has been reached — the field is earlier on the adoption curve than neighbouring environmental applications. Four growth models agree on a three-year forecast of roughly 8,200 articles in 2028 but diverge widely by 2035. Growth is migrating from mapping towards data-driven agronomy and towards actuation, where payloads, precision application and regulation, rather than sensors, are the open problems. The data, the query definitions and the code are provided with this paper so that the analysis can be repeated at any future retrieval date.

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